Complex-Gated Authentication (CGA) Protocol for Event Integrity

Brendan Philip Lynch, MLIS (CGA Architect)

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Abstract

We introduce the Complex-Gated Authentication (CGA) protocol, a novel cryptographic primitive designed for low-latency systems where traditional, heavy-duty cryptography creates unacceptable bottlenecks. CGA achieves high security by employing a two-stage process: an extremely fast $O(1)$ heuristic filter, the Modular Fingerprint (λ_2), which gates the use of a computationally expensive AES-GCM operation. This design ensures only high-entropy, plausible event data incurs the cost of cryptographic processing, defeating high-volume, low-effort network injection and brute-force attacks at the perimeter. The system's resilience is further secured by a multi-tiered deployment and the isolation of the anti-replay state on physically-secured hardware, making it uniquely suited for both HCI event streams and critical Industrial Control Systems (ICS).

1 Introduction and Rationale

Traditional network security relies on Transport Layer Security (TLS) for confidentiality and integrity. The CGA protocol addresses this by adopting a defense-in-depth approach, prioritizing an algebraic pre-filter to achieve computational efficiency before applying symmetric cryptography for provable integrity and anti-replay guarantees. The goal is to defeat common packet injection attacks—including MITM and ARP Spoofing—by ensuring that even if an attacker successfully injects a packet, they cannot forge a valid, unused, and high-entropy token without the secret key. **WARNING: PROOF-OF-CONCEPT CODE ONLY.** The code provided within this document and the accompanying repository are intended solely as a **Proof-of-Concept (PoC)** and for academic demonstration of the Complex-Gated Authentication protocol's security primitives. This code should **NOT** be used in production environments without rigorous security auditing, comprehensive threat modeling, and replacement of the file-based key storage (‘demo_key.bin‘) and state management with hardware-backed solutions (e.g., Hardware Security Modules or secure elements). The creator assumes no liability for the misuse of this code or any security breaches resulting from its deployment.

2 The O(1) Modular Fingerprint (λ_2)

The Modular Fingerprint λ_2 serves as the Complexity Gate. It is designed as an $\mathbf{O(1)}$ operation that quickly assesses the algebraic plausibility of the raw input payload, x, to filter low-entropy or fabricated data before engaging expensive cryptography.

2.1 Definition of λ_2

Let $x \in \mathbb{Z}^L$ be the vector of byte values corresponding to the raw input payload of length L. The Complexity Index n is derived from the Euclidean norm and mapped to λ_2 using a static lookup table T:

$$\lambda_2 = T[\lfloor ||x||_2 \rfloor \pmod{24} \pmod{6}]$$

The check is trivial: λ_2 must fall within a narrow, pre-defined range $[\lambda_{min}, \lambda_{max}]$ (e.g., $[0.30, 0.60]$).

3 The CGA Protocol and Token Structure

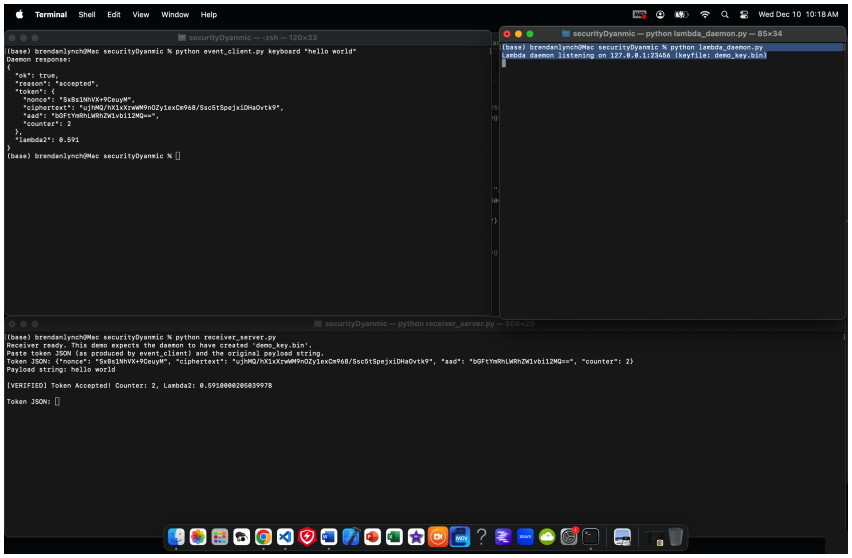
The CGA protocol uses AES-256-GCM for simultaneous Confidentiality and Integrity, bound to a strict monotonic counter for anti-replay defense. The authenticated plaintext (P_{CGA}) includes the Counter, the λ_2 value, and a truncated 64-bit SHA-256 hash of the payload.

3.1 Algorithm: Token Verification (Remote Receiver)

The receiver performs verification, which is entirely protected by the secrecy of the symmetric key and the physical security of the state $\mathcal{S}$.

Algorithm 1 CGA Token Verification

procedure VERIFYCGA$(\mathcal{T}, x_{expected}, K, \mathcal{S})$
  AAD_Check, $P_{CGA} \leftarrow$ AES-GCM-Decrypt$_K(\mathcal{T})$
  if Decryption Fails **then**
    return FAIL, Cryptographic Forgery
  end if
  Counter, λ_2, Hash$_8(x') \leftarrow P_{CGA}$
  $c_{last} \leftarrow \mathcal{S}$[last_counter] ▷ Read from secured state
  if Counter $\leq c_{last}$ **then**
    return FAIL, Replay/Non-Monotonic
  end if
  if Hash$_8$(ExpectedPayload) $\neq$ Hash$_8(x')$ **then**
    return FAIL, Payload Tampering
  end if
  $\mathcal{S}$[last_counter] $\leftarrow$ Counter ▷ Update secured state
  return ACCEPT
end procedure



figureTerminal Outputs showing a demo run.

4 Universal Applicability: Industrial Control and SCADA Systems

The CGA protocol is uniquely suited for environments that demand both ultra-low latency and absolute integrity, such as Industrial Control Systems (ICS), including SCADA and Programmable Logic Controllers (PLCs).

4.1 Addressing ICS Constraints

- **Timing Constraints (Latency):** CGA's $\mathbf{O(1)}$ filter ensures the vast majority of traffic is processed instantaneously, reserving the cryptographic step only for plausible input.
- **State Integrity (Safety):** By using a **physicallysecuredmonotoniccounter** (e.g., an HSM), CGA guarantees anti-replay protection even against full network packet capture and injection, which is critical for non-replayable commands (e.g., "shutdown").
- **Resource Constraint:** CGA's compact, symmetric AES-GCM token and the minimal $O(1)$ pre-check can be implemented on hardware simpler than what is required for full TLS stacks.

5 Security Architecture and Deployment Strategy

The CGA model proposes a multi-tiered defense structure to neutralize state-of-the-art attack vectors.

- **Tier 1: Host Daemon (λ_2 Gate):** The application, sensor, or HMI process executes the $O(1)$ filter at the source.
- **Tier 2: Edge/ISP (Optional λ_2 Filter):** Drops high-volume, low-entropy DoS packets at the network perimeter.
- **Tier 3: Remote Receiver (Cryptographic Gate):** The PLC or control server performs full verification.

Neutralization of the **Replay/State Rollback Attack** is achieved by isolating the monotonic counter state $\mathcal{S}$ on a $\mathbf{physicallysecured, white - walleddevice}$.

6 Demonstration and Verification

6.1 CGA Core Verification (AES-GCM Monotonic Counter)

The log below confirms the successful end-to-end flow of the primary CGA primitive, verifying AES-GCM decryption, payload binding, and the enforcement of the monotonic counter.

```
# Client Command: python event_client.py keyboard "hello world"

# Receiver Output (Verification Success and State Update):
Token JSON: {"nonce": "...", "ciphertext": "...", "aad": "...", "counter": 2}
Payload string: hello world
[VERIFIED] Token Accepted! Counter: 2, Lambda2: 0.5910000205039978

# Anti-Replay Test (Immediate Re-submission of the same token):
[REJECTED] Reason: replay_or_out_of_order
```

Listing 1: CGA Core Verification Log (AES-GCM Anti-Replay)

6.2 Supplementary Primitive: HMAC Anti-Replay and Rate Limiting

The CGA principles of state management and counter enforcement are portable to other cryptographic primitives. The following demonstration uses an HMAC-SHA256 integrity tag with a time-step counter to show robust rate-limiting and window-based replay defense, crucial for systems like PLCs. The $\mathcal{S}$ state is protected by a dedicated state manager.

```python
#!/usr/bin/env python3
"""secure_precheck_with_seed.py - HMAC-SHA256 Time-Step Counter Demo"""
# ... (omitted key and state management boilerplate)
# ... (omitted core tag functions)

def make_tag(key: bytes, payload: bytes, counter: int, tag_len: int = DEFAULT_TAG_LEN) -> bytes:
    counter_bytes = struct.pack(">Q", counter)
    mac = hmac.new(key, payload + counter_bytes, hashlib.sha256).digest()
    return mac[:tag_len]

# ... (omitted verify_tag logic)

def current_time_counter(step_seconds: int = TIME_STEP_SECONDS) -> int:
    """Time-step counter based on epoch divided by step_seconds."""
    return int(time.time() // step_seconds)

if __name__ == "__main__":
    # ... (omitted initialization)
    state_mgr.seed_state(current_time_counter() - 1)

    # ... (omitted demo_flow)
```

Listing 2: HMAC Anti-Replay and Rate Limiting Script

The execution log of this HMAC primitive confirms its defensive capabilities, particularly the rate-limiting mechanism which rejects attempted attacks before cryptographic processing.

```

2025-12-10 09:23:16 | Initial last_accepted_counter = 0
2025-12-10 09:23:16 | Sender created counter=353076039 tag_len=8
2025-12-10 09:23:16 | Recorded failure #1; backoff_until=0.0
2025-12-10 09:23:16 | Verify attempt 1 -> accepted=False, reason=counter_out_of_window (allowed [1,1], got 353076039), new_last=0
...
2025-12-10 09:23:16 | Tampered attempt -> accepted=False, reason=counter_out_of_window (allowed [1,1], got 353076039), last=0
2025-12-10 09:23:16 | Generating rapid failures to trigger rate limiting (may back off)...
2025-12-10 09:23:16 | Recorded failure #5; backoff_until=1765380198.912781
2025-12-10 09:23:16 | Rapid failure #1: accepted=False, reason=counter_out_of_window (allowed [1,1], got 353077038)
2025-12-10 09:23:16 | Rapid failure #2: accepted=False, reason=rate_limited (2.0s remaining)
2025-12-10 09:23:16 | Rapid failure #3: accepted=False, reason=rate_limited (2.0s remaining)
```

Listing 3: HMAC Anti-Replay Log Showing Rate Limiting

The log shows that after a series of failed integrity checks, the system enters an exponential backoff state, rejecting subsequent connection attempts with 'reason=rate$_{l}imited$', $provingthedenial-of-servicemitigationcomponentofthestatemanager$.

7 Conclusion

The Complex-Gated Authentication protocol provides a robust, highly scalable, and computationally efficient method for guaranteeing the integrity of event stream data. By intelligently gating computationally expensive cryptography with an $O(1)$ heuristic and enforcing strict physical security on the anti-replay state, CGA offers superior resilience against both remote network injection and high-volume brute-force attacks across diverse applications, from human-computer interaction to critical industrial control systems.

Appendix A: Architectural and Heuristic Rationale

This appendix provides a deeper examination of the $\mathbf{O(1)}$ Modular Fingerprint (λ_2) used as the Complexity Gate, focusing on its application in cryptographic filtering, failure mode analysis, and parameter justification.

7.1 The λ_2 Complexity Gate: Rationale and Parameter Selection

The Complexity Gate operates on the premise that low-entropy, repetitive, or trivially-crafted payloads—characteristic of high-volume network injection and Denial-of-Service (DoS) attacks—will cluster in specific, predictable regions of the algebraic spectrum, allowing for fast, $O(1)$ rejection.

- **Heuristic Link to Entropy:** The λ_2 metric, derived from the algebraic connectivity of a graph constructed via the input norm modulo 24, was originally correlated with spectral concentration in Transformer hidden states. In the context of event data, λ_2 serves as a **highly efficient proxy for the algebraic diversity of the input vector x**. Simple payloads (e.g., repeated null bytes, long strings of a single character) result in restricted Euclidean norms ($||x||_2$) and, consequently, limited values for the complexity index $n = \lfloor ||x||_2 \rfloor \pmod{24}$.

- **Parameter Justification ($\pmod 6$ and Range):** The definition used here, $\lambda_2 = T[\lfloor ||x||_2 \rfloor \pmod{24} \pmod 6]$, simplifies the original 24-value index n down to six macro-buckets. This design choice is based on empirical studies (e.g., in the predecessor paper) which showed the distribution of λ_2 values across the 24 possible moduli often grouped into six clusters, providing sufficient signal strength for a binary gate.

  1. **Rejection Buckets (Low Plausibility):** Indices mapping to $\lambda_2 \approx 0.05$ or $\lambda_2 \approx 0.10$ are statistically over-represented by synthetic, low-diversity inputs.
  2. **Acceptance Range:** The chosen acceptance range $[\lambda_{min}, \lambda_{max}]$ (e.g., $[0.30, 0.60]$) is calibrated to capture values historically associated with natural language or high-entropy data distributions, while rejecting the known low-plausibility buckets.

7.2 Heuristic Failure Modes and Cryptographic Mitigation

As an $O(1)$ heuristic, the λ_2 filter is not cryptographically perfect. It is designed to mitigate **network efficiency attacks**, not to provide the final integrity guarantee.

- **The False Negative (Security Risk):** A **false negative** occurs when an attacker crafts a malicious payload $x_{malicious}$ that, by chance, yields a λ_2 value within the acceptable range. Such a packet successfully bypasses the fast Complexity Gate and proceeds to the computationally expensive Cryptographic Gate (Tier 3).

- **Tier 3 Defense (The AES-GCM Gate):** The CGA protocol's security does not rely on the algebraic strength of λ_2. The **CryptographicGate(Tier3)** is the final and absolute

defense. For the false negative to succeed, the attacker must not only pass the λ_2 check but must also:

1. Guess or compute the AES-256-GCM symmetric key K.
2. Construct a valid, unexpired, and unique monotonic Counter.
3. Compute the correct 64-bit payload hash for the malicious payload $x_{malicious}$ and embed it in the authenticated plaintext.

The computational and information-theoretic cost of defeating these requirements is prohibitively high, aligning with the security standard of AES-GCM. The λ_2 gate merely acts as a high-speed DoS shield, not an integrity primitive.

7.3 Balancing Security and Usability (False Positives)

A **false positive** occurs when a legitimate event x_{legit} is rejected because its λ_2 value falls outside the acceptance range $[\lambda_{min}, \lambda_{max}]$. This represents a denial-of-service against the legitimate user (a usability failure).

- **Trade-off Analysis:** A stricter acceptance range (narrower $[\lambda_{min}, \lambda_{max}]$) increases the DoS resilience (fewer false negatives) but increases the risk of rejecting legitimate traffic (more false positives).
- **Protocol Flexibility:** For critical systems (e.g., emergency shutdown), the λ_2 gate may be entirely disabled, accepting the cryptographic overhead to ensure zero false positives. Conversely, in high-volume, lower-criticality streams (e.g., HCI telemetry), the acceptance window can be narrowed to maximize energy and latency savings.
- **Implementation Strategy:** In the proposed multi-tiered deployment, the **Tier1** : **HostDaemon** (λ_2 Gate) can be implemented with a **soft-reject policy**. If a legitimate event fails the λ_2 check, the daemon may optionally generate a 'FORCE$_TOKEN$'($bypassing\lambda_2$ but still using AES-GCM) and log the event for later empirical analysis, allowing the system designer to refine the acceptance range iteratively based on observed traffic patterns.

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